

Sunidhi Prakash

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EDUCATION

B.Tech, Indian Institute of Technology Guwahati, India

Jul 2021 – Jun 2025

Bachelors in Engineering Physics

Course Work: Probability Distribution, Computational Physics, Monte Carlo Simulation, Statistics, Machine Learning, Signals & Systems

M.Sc., World Quant University

July 2025 – Present

Masters in Financial Engineering

Course Work: Financial Markets, Financial Data, Financial Econometrics, Derivative Pricing

EXPERIENCE

Axis Bank

June 2025 – Present

Business Analyst - Retail & Deposit

Mumbai, India

- Built **predictive models** and customer analytics using ML techniques like **K-Means clustering & classification** to optimize call scheduling, improve connect rates, and boost cross sell performance by **2%**.
- Built **PySpark** based personalized outbound call-center campaigns using customer data, with real time **eligibility code** automated via **Jenkins & Bitbucket**.
- Analyzed, monitored & backtested campaign performance with **SQL & python** to track outcomes & refine targeting over 7+ campaign variants.

Indian Institute of Technology, Guwahati

May 2024 – Sep 2024

Research Intern under Dr. Sanasam Ranbir Singh in OSINT Lab

Guwahati, India

- Worked on **NLP & multimodal LLMs** (LLaVA, MiniGPT-4, LLaMA-Adapter, Instruct-BLIP) for **sarcasm detection** across **12k** social media posts with **text & images**.
- Finetuned** 4+ models on real world data, **customizing training** workflows to preserve top-performing weights.
- Benchmarked pretrained and finetuned models to **identify limitations** and areas for improvement across **data categories**.

PROJECTS

Large-Scale LP Optimization via Sifting

Feb. 2024 – March. 2024

IISC Bangalore, CISTUP

Github

- Solved a **large-scale LP** problem (**5000 variables, 500 constraints**) using **sifting** method and **python- doCPLEX** module, efficiently reducing it to 896 variables within the 1000-variable limit.
- Built an iterative **dual-primal** approach in **Python & numpy** to find an optimal solution. Built a checker to verify results.

News-Driven Market Risk Indicator Using NLP

Jan. 2026 – Feb. 2026

Personal Project

Github

- Built a financial news **NLP** pipeline to extract **sentiment** and weighted risk factors and integrate them with **S&P 500** data.
- Implemented **ARIMA & ARIMAX** models to evaluate the predictive contribution of news derived risk signals beyond baseline.
- Applied **GARCH** volatility modeling to analyze the relationship between the risk signal & time varying market volatility.

Prompt Engineering in LLMs to solve quantitative problems

Jan. 2024 – April 2024

Guide: Prof. GS Setlur/Department of Physics, IIT Guwahati

Github

- Developed an automated prompt leveraging **ChatGPT, Gemini & Llama APIs** to tackle JEE Advanced problems.
- Used the extensible **LangChain** framework using prompts, **memory, chains**, question answering and **agents**.
- Analyzed model performance across various question types, including single & multiple correct, and numerical questions.

Multi-Resolution STFT Analysis

April. 2025 – May. 2025

IIT Guwahati: Course Project

Github

- Developed a **multi-window STFT** approach in **Python** to capture both transient and sustained audio features, overcoming **time-frequency** trade-offs
- Tested on tonal and percussive sounds, showing improved temporal and spectral representation over **Constant-Q Transform**.

TECHNICAL SKILLS

- Statistics & Modelling** : Time Series Forecasting, Regression, Hypothesis Testing
- ML Frameworks**: Transformer, Pytorch, Hugging Face, Tensorflow, Keras, Langchain
- Programming**: C/C++, Python, R
- Data Processing**: Pandas, Hadoop, PowerBI, AWS
- Database Management**: PostgreSQL, MySQL

CERTIFICATION

- Algorithms Specialization**: Stanford, Coursera
- PostgreSQL Specialization**: University of Michigan, Coursera
- Machine Learning & Deep Learning Specialization**: Stanford
- Introduction to Big Data with Spark & Hadoop**: IBM
- Bayesian Statistics & Time Series Analysis**: UC Santa Cruz, Coursera